

Experience

Apple ▪ Senior Software Developer (Contract)

Video Computer Vision Team, Machine Learning and AI Group March 2024 – Present

- Productionized research prototypes for Apple foundation model optimization using Vue, Svelte, and Flask
- Led design and development of internal Next.js storefronts for configuring, integrating, and sharing LLMs
- Expanded interactive graph visualization tool Mycelium (open-source) supporting on-device optimization tools
- Maintained and improved TypeScript libraries for visualizing deep neural networks and architectures

Memorial Sloan Kettering Cancer Center ▪ Senior Software Engineer

Pathology Results Team, Clinical Bioinformatics Group May 2018 – Present

- Architected and implemented performant web apps for managing and visualizing high-throughput clinical data supporting 100k+ patients, using JavaScript, React, Python, Flask, MySQL, and MongoDB
- Redesigned and rewrote core UIs, improving query times, performance metrics, code splitting, end-to-end testing, state machines, and application monitoring, achieving 20% faster clinical report turnaround times
- Built internal JavaScript framework, design system, and CI/CD pipeline for teams of 30+ developers to reduce code duplication, standardize tooling, and establish visual unity across 10+ web portals
- Developed ETL systems and REST APIs with 2x faster speeds in production using Python, Airflow, and k8s

Q3P ▪ Software Consultant (Freelance)

Music Artist Activations Team, Various Record Labels Dec 2022 – Present

- Created web platform used to manage digital communities for premier music artists, serving 1M+ daily active users

Skills

Languages ▪ JavaScript, TypeScript, Python, Ruby, SQL, HTML, CSS, Sass

Libraries ▪ React, Flask, FastAPI, Next.js, Vue, Node.js, pandas, Playwright, Cypress

Tools ▪ Git, Docker, GitLab CI/CD, GitHub Actions, Apache Airflow, nginx, AWS S3, ELK, k8s

Design ▪ Figma, Adobe Photoshop, Lightroom, Illustrator

Education

Macaulay Honors College New York, NY

B.A. Chemistry and Bioinformatics, Minor in Mathematics, GPA 3.95 Aug 2013 – May 2017

- Data Science Fellowship 2016
- Full Merit Scholarship 2013 – 2017

Publications

Khwaja E, Song YS, **Agarunov A**, Huang B. CELL-E 2: Translating Proteins to Pictures and Back with a Bidirectional Text-to-Image Transformer. **NeurIPS** (2023).

Brannon AR, [et al, including **Agarunov A**]. Enhanced specificity of clinical high-sensitivity tumor mutation profiling in cfDNA via paired normal sequencing using MSK-ACCESS. **Nature Communications**. 12, 3934 (2021).

Projects

Turbo Starter ▪ Template to build monorepo microfrontend React apps with testing, tooling, docs 70 GitHub ★

kume ▪ Interactive visualization of color quantization using k-means clustering

ronivonu ▪ Generative art created in JavaScript, Python, and Processing